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Data Transmission Method and Apparatus

Abstract

A data transmission method includes establishing a first communication link with an internet of things platform; sending a multi-link access request to the internet of things platform over the first communication link, where the multi-link access request carries a first quantity of internet of things devices accessing an edge gateway; receiving link establishment information that is sent by the internet of things platform and that corresponds to at least one second communication link; establishing the at least one second communication link with the internet of things platform based on the link establishment information corresponding to the at least one second communication link; and performing data transmission with the internet of things platform over the first communication link and the second communication link.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a continuation of International Patent Application No. PCT/CN2023/104842 filed on Jun. 30, 2023, which claims priority to Chinese Patent Application No. 202211262656.1 filed on Oct. 14, 2022, and Chinese Patent Application No. 202211734961.6 filed on Dec. 30, 2022. All of the aforementioned patent applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

[0002] This application relates to the field of internet of things technologies, and in particular, to a data transmission method and an apparatus.

BACKGROUND

[0003] In an internet of things system, it is often necessary to connect various internet of things devices. However, some internet of things devices lacking a network communication capability need to access an internet of things platform via an edge gateway.

[0004] With the enhancement of a computing capability of the edge gateway, a single edge gateway may be accessed by a large quantity of internet of things devices. In this case, a communication link between the edge gateway and the internet of things platform needs to carry a large amount of internet of things device data. This may cause congestion on the communication link.

SUMMARY

[0005] Embodiments of this application provide a data transmission method and an apparatus to avoid congestion on a communication link between an edge gateway and an internet of things platform. The technical solutions are as follows.

[0006] According to a first aspect, a data transmission method is provided. The method is applied to an edge gateway, and the method includes establishing a first communication link with an internet of things platform; sending a multi-link access request to the internet of things platform over the first communication link, where the multi-link access request carries a first quantity of internet of things devices accessing the edge gateway; receiving link establishment information that is sent by the internet of things platform and that corresponds to at least one second communication link; establishing the at least one second communication link with the internet of things platform based on the link establishment information corresponding to the at least one second communication link; and performing data transmission with the internet of things platform over the first communication link and the second communication link.

[0007] In the solution shown in this application, the edge gateway may first establish the first communication link with the internet of things platform, and then send the multi-link access request to the internet of things platform over the first communication link, to subsequently establish the at least one second communication link with the internet of things platform. In this way, data may be transmitted between the edge gateway and the internet of things platform over a plurality of communication links (the first communication link and the at least one second communication link), so that a probability of congestion on the communication link can be reduced.

[0008] In an implementation, establishing the at least one second communication link with the internet of things platform based on the link establishment information corresponding to the at least

one second communication link includes sending, to the internet of things platform based on the link establishment information corresponding to the at least one second communication link, a link establishment request respectively corresponding to the at least one second communication link, where the link establishment request carries a gateway identifier of the edge gateway, the gateway identifier is different from a gateway identifier used by the edge gateway to establish the first communication link, and when there are a plurality of link establishment requests, each link establishment request carries a different gateway identifier.

[0009] In the solution shown in this application, the edge gateway may generate a plurality of different gateway identifiers based on the gateway identifier of the edge gateway, and gateway identifiers carried in link establishment requests that are sent by the edge gateway to the internet of things platform and that correspond to the second communication links are different. In this way, the plurality of communication links can be established with the internet of things platform based on different gateway identifiers, so that a probability of congestion on the communication link can be reduced.

[0010] In an implementation, the link establishment information includes a private port number of a service instance corresponding to the second communication link. The establishing the at least one second communication link with the internet of things platform based on the link establishment information corresponding to the at least one second communication link includes sending, to the internet of things platform, the link establishment request respectively corresponding to the at least one second communication link, where the link establishment request carries the private port number corresponding to the second communication link.

[0011] In the solution shown in this application, a port number corresponding to the service instance is the private port number. In this way, the internet of things platform can establish the second communication link with a specified service instance based on the private port number, so that load balancing of the service instance can be implemented.

[0012] In an implementation, performing data transmission with the internet of things platform over the first communication link and the second communication link includes selecting a target communication link from the first communication link and the second communication link according to a specified communication link selection policy each time data needs to be sent to the internet of things platform; and sending the data over the target communication link.

[0013] In the solution shown in this application, the edge gateway may select, from the first communication link and the second communication link according to the communication link selection policy, the target communication link for sending the data to the internet of things platform. In this way, data originally sent by one communication link can be sequentially shared to different communication links for sending, so that a probability of congestion on the communication link can be reduced.

[0014] In an implementation, the communication link selection policy includes random selection or sequential selection.

[0015] In an implementation, the multi-link access request further carries a communication frequency of the internet of things device.

[0016] In the solution shown in this application, the multi-link access request sent by the edge gateway to the internet of things platform may further include the communication frequency. In this way, a service instance on the internet of things platform may determine, based on the first quantity of internet of things devices accessing the edge gateway and the communication frequency of the internet of things device, an amount of data for communication between the edge gateway and the internet of things platform, and the internet of things platform may determine, based on the amount of data, a quantity of second communication links to be established with the edge gateway. In this way, dynamic allocation of the communication links can be implemented, and a probability of congestion on the communication links between the edge gateway and the internet of things platform can be reduced.

[0017] According to a second aspect, a data transmission method is provided. The method is applied to an internet of things platform, and the method includes receiving a multi-link access request sent by an edge gateway over a first communication link, where the multi-link access request carries a first quantity of internet of things devices accessing the edge gateway; determining, based on the first quantity, a second quantity of second communication links to be established with the edge gateway; obtaining link establishment information corresponding to the second quantity of second communication links; sending, to the edge gateway, the link establishment information corresponding to the second quantity of second communication links, where the link establishment information indicates the edge gateway to establish the second quantity of second communication links with the internet of things platform; and performing data transmission with the edge gateway over the first communication link and the second communication link.

[0018] In the solution shown in this application, after establishing the first communication link with the edge gateway, the internet of things platform may receive the multi-link access request that is sent by the edge gateway over the first communication link and that carries the first quantity of internet of things devices accessing the edge gateway. Then, the internet of things platform may determine, based on the first quantity of internet of things devices of the edge gateway, the second quantity of second communication links to be established with the edge gateway, and then send, to the edge gateway, the link establishment information corresponding to the second quantity of second communication links, to subsequently establish the second quantity of second communication links with the edge gateway. In this way, data may be transmitted between the internet of things platform and the edge gateway over a plurality of communication links (the first communication link and at least one second communication link), so that a probability of congestion on the communication link can be reduced.

[0019] In an implementation, determining, based on the first quantity, a second quantity of second communication links to be established with the edge gateway includes determining, based on a first correspondence, the second quantity that corresponds to the first quantity and that is of the second communication links to be established with the edge gateway.

[0020] In the solution shown in this application, a correspondence between the first quantity of internet of things devices accessing the edge gateway and the second quantity of communication links subsequently established between the internet of things platform and the edge gateway may be set on the internet of things platform. In the correspondence, the first quantity is positively correlated with the second quantity. For example, a larger quantity of internet of things devices accessing the edge gateway indicates a larger quantity of communication links established between the edge gateway and the internet of things platform. In this way, a quantity of communication links established between the edge gateway and the internet of things platform can be dynamically determined based on the first quantity of internet of things devices accessing the edge gateway, thereby avoiding a waste of resources.

[0021] In an implementation, the multi-link access request further carries a communication frequency of the internet of things device accessing the edge gateway. The determining, based on the first quantity, a second quantity of second communication links to be established with the edge gateway includes determining, based on the first quantity and the communication frequency, the second quantity of second communication links to be established with the edge gateway.

[0022] In an implementation, determining, based on the first quantity and the communication frequency, the second quantity of second communication links to be established with the edge gateway includes determining a product of the first quantity and the communication frequency; and determining, based on a second correspondence, the second quantity that corresponds to the product and that is of the second communication links to be established with the edge gateway.

[0023] In the solution shown in this application, the multi-link access request sent by the edge gateway to the internet of things platform may further include the communication frequency. In this

way, a service instance on the internet of things platform may determine, based on the first quantity of internet of things devices accessing the edge gateway and the communication frequency of the internet of things device, an amount of data for communication between the edge gateway and the internet of things platform, and the internet of things platform may determine, based on the amount of data, a quantity of second communication links to be established with the edge gateway. In this way, dynamic allocation of communication links can be implemented, a probability of congestion on the communication link between the edge gateway and the internet of things platform can be reduced, and a waste of resources can be avoided.

[0024] In an implementation, obtaining link establishment information corresponding to the second quantity of second communication links includes determining, from service instances other than a service instance corresponding to the first communication link, service instances of the second quantity with least corresponding load; and obtaining link establishment information respectively corresponding to the second quantity of service instances.

[0025] In the solution shown in this application, when a plurality of service instances for which the edge gateway establishes a link are determined, the second quantity of service instances with least corresponding load may be determined as a service instance for establishing the second communication link with the edge gateway. In this way, load balancing between the service instances can be implemented.

[0026] In an implementation, the link establishment information includes a private port number corresponding to a service instance. Before performing the data transmission with the edge gateway over the first communication link and the second communication link, the method further includes receiving a link establishment request that is sent by the edge gateway and that carries a private port number; and establishing a second communication link between the edge gateway and a service instance corresponding to the private port number.

[0027] In the solution shown in this application, a port number corresponding to the service instance may be set to the private port number. In this way, the internet of things platform can establish the second communication link between the edge gateway and a specified service instance based on the private port number carried in the link establishment information sent by the edge gateway, so that load balancing of the service instance can be implemented.

[0028] In an implementation, performing the data transmission with the edge gateway over the first communication link and the second communication link includes selecting a target communication link from the first communication link and the second communication link according to a specified communication link selection policy each time data needs to be sent to the edge gateway; and sending the data over the target communication link.

[0029] In an implementation, the communication link selection policy includes random selection or sequential selection.

[0030] In the solution shown in this application, the internet of things platform may select, from the first communication link and the second communication link according to the communication link selection policy, the target communication link for sending the data to the edge gateway. In this way, data originally sent by one communication link can be sequentially shared to different communication links for sending, so that a probability of congestion on the communication link can be reduced.

[0031] According to a third aspect, a data transmission apparatus is provided. The apparatus includes at least one module configured to implement the data transmission method provided in any one of the first aspect or the implementations of the first aspect.

[0032] According to a fourth aspect, a data transmission apparatus is provided. The apparatus includes at least one module configured to implement the data transmission method provided in any one of the second aspect or the implementations of the second aspect.

[0033] According to a fifth aspect, a data transmission system is provided. The system includes an edge gateway and an internet of things platform. The edge gateway is configured to implement the

data transmission method provided in any one of the first aspect or the implementations of the first aspect. The internet of things platform implements the data transmission method provided in any one of the second aspect or the implementations of the second aspect.

[0034] According to a sixth aspect, an edge gateway is provided. The edge gateway includes a processor and a memory. The memory stores at least one instruction. The at least one instruction is loaded and executed by the processor to implement the data transmission method provided in any one of the first aspect or the examples of the first aspect.

[0035] According to a seventh aspect, a computing device cluster is provided, and includes at least one computing device. Each computing device includes a processor and a memory. The processor of the at least one computing device is configured to execute instructions stored in the memory of the at least one computing device, to enable the computing device cluster to perform the data transmission method provided in any one of the second aspect or the examples of the second aspect.

[0036] According to an eighth aspect, a computer program product including instructions is provided. When the instructions are run by an edge gateway, the edge gateway is enabled to perform the data transmission method provided in any one of the first aspect or the examples of the first aspect.

[0037] According to a ninth aspect, a computer program product including instructions is provided. When the instructions are run by a computing device cluster, the computing device cluster is enabled to perform the data transmission method provided in any one of the second aspect or the examples of the second aspect.

[0038] According to a tenth aspect, a computer-readable storage medium is provided, and includes computer program instructions. When the computer program instructions are executed by an edge gateway, the edge gateway performs the data transmission method provided in any one of the first aspect or the examples of the first aspect.

[0039] According to an eleventh aspect, a computer-readable storage medium is provided, and includes computer program instructions. When the computer program instructions are executed by a computing device cluster, the computing device cluster performs the data transmission method provided in the second aspect or any one of the examples of the second aspect.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0040] FIG. 1 is a diagram of a system architecture of an internet of things system according to this application;

[0041] FIG. 2 is a flowchart of a data transmission method according to this application;

[0042] FIG. 3 is a flowchart of a data transmission method according to this application;

[0043] FIG. 4A and FIG. 4B are a flowchart of a method for establishing a plurality of links between an edge gateway and a server according to this application;

[0044] FIG. 5 is a flowchart of a method for reporting data by an edge gateway according to this application;

[0045] FIG. 6 is a flowchart of a method for sending a message to an edge network by an internet of things platform according to this application;

[0046] FIG. 7 is a flowchart of a method for disconnecting a connection to an edge gateway in a multi-link mode according to this application;

[0047] FIG. 8 is a diagram of a structure of a data transmission apparatus according to this application;

[0048] FIG. 9 is a diagram of a structure of a data transmission apparatus according to this application;

[0049] FIG. **10** is a diagram of a computing device according to this application;
[0050] FIG. **11** is a diagram of a computing device cluster according to this application; and
[0051] FIG. **12** is a connection diagram of a computing device according to this application.

DESCRIPTION OF EMBODIMENTS

[0052] To make the objectives, technical solutions, and advantages of this application clearer, the following further describes implementations of this application in detail with reference to the accompanying drawings.

[0053] FIG. **1** is a diagram of a system architecture of an internet of things system according to this application. Refer to FIG. **1**. The internet of things system includes internet of things devices **101**, an edge gateway **102**, and an internet of things platform **103**.

[0054] The internet of things devices **101** may be various smart homes (such as smart refrigerators and smart air conditioners), various industrial devices (such as smart water meters and smart electricity meters), various intelligent sensors (such as smoke sensors and temperature and humidity sensors), or the like. The edge gateway **102** has a function like data exchange, and may be a router, a switch, or the like. The internet of things platform **103** may be an internet of things server. The internet of things server may be an independent server, or a server cluster. One or more internet of things devices **101** may access the edge gateway **102**, and may establish a communication link with the internet of things platform **103** via the edge gateway **102**, to perform data transmission.

[0055] A plurality of service instances is disposed in the internet of things platform **103**, and each of the service instances may be a program set that is run in a server and that can independently provide a service for the internet of things device. The plurality of service instances includes at least a load balancing service, an access service, and a service processing service. In related technologies, the edge gateway may first send a link establishment request to the load balancing service, and the load balancing service may be one access service selected from a plurality of access services, to establish, with the edge gateway, a communication link for data transmission.

[0056] After the edge gateway establishes the communication link with the access service on the internet of things platform, the internet of things device accessing the edge gateway may send data (for example, collected industrial data) to the access service on the internet of things platform over the communication link. After receiving the data sent by the internet of things device, the access service may forward corresponding data to a corresponding service processing service, and the corresponding service processing service processes the data. The service processing service may alternatively send data (for example, a control instruction for an industrial device) to the edge gateway by using the access service.

[0057] With the enhancement of a computing capability of the edge gateway, a single edge gateway may be accessed by a large quantity of internet of things devices. In this case, data transmission is performed between the edge gateway and the internet of things platform over only one communication link. This may cause congestion on the communication link.

[0058] This application provides a data transmission method, to avoid congestion on a transmission link between an edge gateway and an internet of things platform. The following describes the data transmission method provided in this application with reference to the accompanying drawings.

[0059] FIG. **2** shows a data transmission method according to this application. The method may be applied to an edge gateway. Refer to FIG. **2**. The method includes the following steps.

[0060] Step **201**: The edge gateway establishes a first communication link with an internet of things platform.

[0061] The first communication link is a 1.sup.st communication link established between the edge gateway and a service instance corresponding to an access service on the internet of things platform. For establishment of the first communication link, refer to other technologies. Details are not described herein. The service instance accessing the edge gateway over the first communication link may be subsequently referred to as a first service instance.

[0062] Step **202**: Send a multi-link access request to the internet of things platform over the first communication link, where the multi-link access request carries a first quantity of internet of things devices accessing the edge gateway.

[0063] After establishing the first communication link with the service instance on the internet of things platform, if determining that a multi-link establishment condition is currently met, the edge gateway may send, over the first communication link, the multi-link access request to the first service instance connected to the first communication link. The multi-link establishment condition may be that the first quantity of internet of things devices accessing the edge gateway is greater than a threshold of a quantity of internet of things devices.

[0064] The multi-link access request sent to the internet of things platform carries the first quantity of internet of things devices accessing the edge gateway, so that the internet of things platform determines, based on the first quantity of internet of things devices accessing the edge gateway, a second quantity of second communication links to be established subsequently with the edge gateway. The second communication link is a communication link established between the edge gateway and a service instance other than the first service instance on the internet of things platform. The service instance accessing the edge gateway over the second communication link may be subsequently referred to as a second service instance.

[0065] The multi-link access request sent to the internet of things platform may further carry a communication frequency of the internet of things device. The communication frequency is a frequency at which the internet of things device transmits data to the internet of things platform. The internet of things platform may determine, based on the first quantity of internet of things devices accessing the edge gateway and the communication frequency of the internet of things device, a quantity of second communication links to be established subsequently with the edge gateway.

[0066] Step **203**: Receive link establishment information that is sent by the internet of things platform and that corresponds to at least one second communication link.

[0067] After receiving the multi-link access request sent by the edge gateway, the service instance that corresponds to the first communication link and that is on the internet of things platform may determine, based on the multi-link access request, the link establishment information corresponding to the at least one second communication link, and return the link establishment information to the edge gateway. A quantity of second communication links is the same as a quantity of second service instances.

[0068] Step **204**: Establish the at least one second communication link with the internet of things platform based on the link establishment information corresponding to the at least one second communication link.

[0069] After receiving the link establishment information corresponding to the at least one second communication link, the edge gateway may send a link establishment request to the internet of things platform based on link establishment information corresponding to each second communication link, to establish a communication link with the second service instance on the internet of things platform.

[0070] In an example, gateway identifiers carried in a plurality of link establishment requests sent by the edge gateway to the internet of things platform are different. The plurality of gateway identifiers each are a gateway identifier corresponding to the edge gateway. The plurality of link establishment requests includes a link establishment request sent when the edge gateway establishes the first communication link with the internet of things platform and a link establishment request sent when the edge gateway establishes the second communication link with the internet of things platform. For example, the edge gateway may generate different gateway identifiers based on an original gateway identifier of the edge gateway and a quantity of times of sending a link establishment request to the internet of things platform currently.

[0071] For example, the gateway identifier of the edge gateway is GW001. When the edge gateway

sends a link establishment request to the internet of things platform for the first time, a gateway identifier carried in the link establishment request is GW001-1. When the edge gateway sends a link establishment request to the internet of things platform for the second time, a gateway identifier carried in the link establishment request is GW001-2. When the edge gateway sends a link establishment request to the internet of things platform for the N.sup.th time, a gateway identifier carried in the link establishment request is GW001-N. In this way, link establishment requests that are separately received by the internet of things platform and that carry different gateway identifiers may be considered as being sent by different edge gateways, so that the plurality of link establishment requests can be prevented from being processed as a same link establishment request because the gateway identifiers carried in the plurality of link establishment requests are the same. The internet of things platform may separately verify different link establishment requests, to establish a plurality of second communication links with the edge gateway.

[0072] In addition, the link establishment information corresponding to the second communication link further includes a port number of the second service instance. In an example, the port number of the second service instance is a private port number. In this way, the link establishment request sent by the edge gateway to the internet of things platform may carry the private port number corresponding to the second service instance, to establish the second communication link with the specified second service instance on the internet of things platform.

[0073] Step **205**: Perform data transmission with the internet of things platform over the first communication link and the second communication link.

[0074] After establishing the first communication link and the second communication link with the internet of things platform, the edge gateway may perform data transmission with the internet of things platform over the first communication link and the second communication link.

[0075] Each time the edge gateway needs to send data to the internet of things platform, the edge gateway may select a target communication link from the first communication link and the second communication link according to a specified communication link selection policy, and send the data to the internet of things platform over the selected target communication link. The communication link selection policy includes random selection or sequential selection. In addition, when disconnecting the communication link from the internet of things platform, the edge gateway may send a link disconnection notification to the internet of things platform for a plurality of communication links (the first communication link and the second communication link) established with the internet of things platform, to disconnect the plurality of communication links established with the internet of things platform.

[0076] According to this embodiment provided in this application, the plurality of communication links may be established between the edge gateway and the internet of things platform for data transmission, and each communication link may be used for data transmission with a different service instance on the internet of things platform. In this way, a communication link between the edge gateway and the internet of things platform is increased, thereby reducing pressure of data transmission over each communication link, and reducing a probability of congestion on the communication link.

[0077] FIG. 3 shows a data transmission method according to this application. The method may be applied to an internet of things platform. Refer to FIG. 3, the method includes the following steps.

[0078] Step **301**: The internet of things platform receives a multi-link access request sent by an edge gateway over a first communication link, where the multi-link access request carries a first quantity of internet of things devices accessing the edge gateway.

[0079] The first communication link is a 1.sup.st communication link established between the edge gateway and a service instance on the internet of things platform. For establishment of the first communication link, refer to other technologies. Details are not described herein. The service instance accessing the edge gateway over the first communication link may be subsequently

referred to as a first service instance. Before step **301** is performed, the first communication link may be first established between the edge gateway and the first service instance in the internet of things platform.

[0080] Step **302**: Determine, based on the first quantity, a second quantity of second communication links to be established with the edge gateway.

[0081] When the multi-link access request sent by the edge gateway carries the first quantity of internet of things devices accessing the edge gateway, the first service instance may determine, based on the first quantity of internet of things devices accessing the edge gateway, the second quantity of communication links subsequently established by the internet of things platform with the edge gateway. The communication link subsequently established by the internet of things platform with the edge gateway may be referred to as a second communication link.

[0082] In an example, the internet of things platform may store a first correspondence between a quantity of internet of things devices and a quantity of communication links. For the second quantity of communication links subsequently established by the internet of things platform with the edge gateway, the first service instance may determine, by searching for the first correspondence, the second quantity that corresponds to the first quantity and that is of the second communication links to be established with the edge gateway.

[0083] In another example, the multi-link access request further carries a communication frequency of the internet of things device accessing the edge gateway, and the communication frequency is a frequency at which the internet of things device transmits data to the internet of things platform. The first service instance may calculate a product of the communication frequency and the first quantity, to obtain a total communication frequency corresponding to the edge gateway. The internet of things platform may store a second correspondence between the communication frequency of the edge gateway and a quantity of communication links. The first service instance may determine, by searching for the second correspondence, the second quantity that corresponds to the total communication frequency and that is of the second communication links to be established with the edge gateway.

[0084] Step **303**: Obtain link establishment information corresponding to the second quantity of second communication links.

[0085] After determining the second quantity of second communication links to be established with the edge gateway, the first service instance may obtain the link establishment information corresponding to the second quantity of second communication links.

[0086] The link establishment information corresponding to the second communication link may be link establishment information corresponding to a service instance other than the first service instance on the internet of things platform, and the service instance is a service instance corresponding to an access service. The service instance may be referred to as a second service instance. The link establishment information corresponding to the second communication link may include a port number corresponding to the second service instance.

[0087] In an example, the port number corresponding to the second service instance may be a private port number. In this way, after receiving the link establishment information, the edge gateway may send a link establishment request carrying the private port number included in the link establishment information to a load balancing service of the internet of things platform. The load balancing service may send, based on the private port number carried in the link establishment request, the corresponding link establishment request to the second service instance corresponding to the private port number, to establish the second communication link between the edge gateway and the second service instance.

[0088] In an example, the second quantity of second service instances may be a second quantity of service instances with least corresponding load other than the first service instance in the internet of things platform.

[0089] After determining the second quantity of second communication links to be established

subsequently with the edge gateway, the first service instance may obtain load information corresponding to each second service instance, determine, from a plurality of second service instances based on the load information, service instances of the second quantity with least corresponding load, and then determine the second quantity of service instances with least load as service instances that provide access services for the edge gateway.

[0090] Step **304**: Send the link establishment information corresponding to the second quantity of second communication links to the edge gateway.

[0091] Step **305**: Perform data transmission with the edge gateway over the first communication link and the second communication link.

[0092] After the edge gateway establishes the first communication link and the second communication link with the internet of things platform, the internet of things platform may perform data transmission with the edge gateway over the first communication link and the second communication link.

[0093] Each time the internet of things platform needs to send data to the edge gateway, the internet of things platform may select a target communication link from the first communication link and the second communication link according to a specified communication link selection policy, and send the data to the edge gateway over the selected target communication link. The communication link selection policy includes random selection or sequential selection.

[0094] According to this embodiment provided in this application, the plurality of communication links may be established between the edge gateway and the internet of things platform for data transmission, and each communication link may be used for data transmission with a different service instance on the internet of things platform. In this way, a communication link between the edge gateway and the internet of things platform is increased, thereby reducing pressure of data transmission over each communication link, and reducing a probability of congestion on the communication link.

[0095] In this application, a communication link may be established between the edge gateway and the service instance on the internet of things platform according to a message queue telemetry transport (MQTT) protocol. A plurality of ports in the internet of things platform are opened for an MQTT interface used for accessing the edge gateway, and configuration of a routing distribution policy for a port is completed on a load balancing (LB) service. The configuration of the routing distribution policy is as follows.

[0096] For one well-known port (for example, 8883), that is, a well-known port of an MQTT interface provided externally, the edge gateway may initially access the LB of the internet of things platform through the well-known port. The routing distribution policy configured on the LB is that when a link establishment request carrying a well-known port number is received, corresponding link establishment requests are distributed in an access service cluster at a backend of the internet of things platform in a polling mode.

[0097] For a plurality of private ports (for example, 8981, 8982, 8983, and so on), a quantity of private ports is consistent with a quantity of access services at the backend of the internet of things platform. The routing distribution policy configured on the LB is that when a link establishment request carrying a private port number is received, the corresponding link establishment request is sent to the access service corresponding to the private port number.

[0098] FIG. 4A and FIG. 4B are a flowchart of a method for establishing a plurality of communication links between an edge gateway and an internet of things platform according to this application. Refer to FIG. 4A and FIG. 4B, the method includes the following steps.

[0099] Step **401**: The edge gateway performs initial configuration. The edge gateway completes configuration of a key parameter. The key parameter includes a user name and a password that are used on the internet of things platform for authentication, and a server address and a well-known port number of a load service on the internet of things platform.

[0100] Step **402**: The internet of things platform performs initial configuration. The internet of

things platform completes configuration of a routing distribution policy for load balancing based on a service deployment status of the internet of things platform.

[0101] Step **403**: The edge gateway initiates a link establishment request. The edge gateway sends the link establishment request to the LB based on the server address and a well-known port of the load service, to complete link establishment with the load service.

[0102] Step **404**: The LB performs routing distribution for the well-known port number carried in the link establishment request. The LB performs distribution for the well-known port according to a polling policy, and selects a first service instance (for example, a service instance 1) from an access service cluster at a backend, to complete link establishment between the LB and the first service instance.

[0103] Step **405**: The edge gateway sends an MQTT CONNECT message to the LB. The edge gateway sends the MQTT CONNECT message to the internet of things platform through a standard MQTT interface. The MQTT CONNECT message carries basic identity information (a device Id) of the edge gateway and authentication information (a user name and a password) of the edge gateway.

[0104] Step **406**: The LB performs message distribution. The LB sends the MQTT CONNECT message to the first service instance over a link established with the first service instance.

[0105] Step **407**: The first service instance completes authentication and performs connection binding. The first service instance performs authentication on the authentication information carried in the MQTT CONNECT message, and performs connection binding. The first service instance is bound to the edge gateway as an available route.

[0106] Step **408**: The first service instance returns an authentication success.

[0107] Step **409**: The LB returns, over the link that is already established with the edge gateway, the authentication success. In this case, a first communication link is successfully established between the edge gateway and the first service instance.

[0108] Step **410**: The edge gateway sends an MQTT PUBLISH message (that is, a multi-link access request) to the first service instance: The edge gateway determines, based on a quantity of subdevices accessing the edge gateway, whether to send the MQTT PUBLISH message to the first service instance. If the quantity of subdevices accessing the edge gateway exceeds a specified quantity threshold, it may be considered that the edge gateway is a large gateway, and multi-link access may be applied for to the internet of things platform.

[0109] An example of the MQTT PUBLISH message is as follows: [0110] topic:

\$oc/devices/{device Id}/sys/links/request; [0111] payload: {"devices_number": 10000,

"Frequency": 60} [0112] device Id is an identifier of the edge gateway, devices_number is the quantity of subdevices accessing the edge gateway, and Frequency is a communication frequency of the subdevice.

[0113] Step **411**: The first service instance responds to the MQTT PUBLISH message. The first service instance determines, based on the quantity of subdevices that is carried in the MQTT PUBLISH and message the communication frequency, a second quantity of communication links to be established subsequently with the edge gateway; determines external link establishment information of a second quantity of second service instances based on a current actual load status of another service instance; and sends the link establishment information to the edge gateway via an MQTT message.

[0114] An example of the MQTT message is as follows: [0115] topic:

\$oc/devices/{device_id}/sys/links/response; [0116] payload: {"services_num": 2, "list":

[{"client_id": "id1", port: 8902}, {"client_id": "id2", port: 8903}]} [0117] client_id is an instance identifier of the second service instance, and port is a private port number corresponding to the second service instance.

[0118] Step **412**: The edge gateway sends an MQTT CONNECT message to the LB. The edge gateway establishes a link with the second service instance based on the link establishment

information obtained via the request, for example, establishes a link from the port **8902**.

[0119] The MQTT CONNECT message carries the basic identity information of the edge gateway, the authentication information of the edge gateway, and an instance identifier and a port number (**8902**) that correspond to the second service instance.

[0120] Step **413**: LB performs routing distribution. The LB routes the MQTT CONNECT message to the second service instance (for example, a service instance **2**) according to a fixed distribution policy for the port **8902**.

[0121] Step **414**: The service instance **2** completes authentication and performs connection binding. Access authentication is completed based on specified client_id, establishment of a second communication link is completed, and the connection binding is performed. The service instance **2** is bound to the edge gateway as an available route.

[0122] In this case, the second communication link is successfully established between the edge gateway and the service instance **2**. Such as, currently, the first communication link is established between the edge gateway and the service instance **1**, and the second communication link is successfully established between the edge gateway and the service instance **2**.

[0123] Step **415**: The edge gateway sends an MQTT CONNECT message to the LB. The edge gateway establishes a link with the second service instance based on the link establishment information obtained via the request, for example, establishes a link from the port **8903**.

[0124] Step **416**: LB performs routing distribution. The LB routes the MQTT CONNECT message to the second service instance (for example, a service instance **3**) according to a fixed distribution policy for the port **8903**.

[0125] Step **417**: The service instance **3** performs authentication and connection binding. Access authentication is completed based on specified client_id, establishment of the second communication link is completed, and the connection binding is performed. The service instance **3** is bound to the edge gateway as an available route.

[0126] In this case, the second communication link is successfully established between the edge gateway and the service instance **3**. In a possible design, currently, the first communication link is established between the edge gateway and the service instance **1**, the second communication link is successfully established between the edge gateway and the service instance **2**, and the second communication link is successfully established between the edge gateway and the service instance **3**. In this way, the edge gateway may perform data transmission with the internet of things platform over three communication links, and data on each communication link is processed by different service instances, so that congestion on the communication link between the edge gateway and the internet of things platform can be avoided.

[0127] FIG. 5 is a flowchart of a method for reporting data by an edge gateway to an internet of things platform according to this application. Based on multi-link uplink communication, the edge gateway actively sends a message (for example, reports collected data sent by an internet of things device) to the internet of things platform. Refer to FIG. 5, the method includes the following steps.

[0128] Step **501**: The edge gateway reports data to the internet of things platform based on a service scenario.

[0129] Step **502**: The edge gateway randomly selects a communication link (for example, selects a communication link **1**) to report the data.

[0130] Step **503**: LB routes, through a corresponding connection, to a service instance **1** that corresponds to the communication link **1** and that is at a backend of the internet of things platform.

[0131] Step **504**: A backend service corresponding to the service instance **1** normally processes a service request.

[0132] Step **505**: The edge gateway reports data to the internet of things platform based on a service scenario.

[0133] Step **506**: The edge gateway randomly selects a communication link (for example, selects a communication link **2**) to report the data.

[0134] Step **507**: The LB routes, through a corresponding connection, to a service instance **2** that corresponds to the communication link **2** and that is at a backend of the internet of things platform.

[0135] Step **508**: A backend service corresponding to the service instance **2** normally processes a service request.

[0136] In the method for reporting data by the edge gateway to the internet of things platform provided in this embodiment of this application, the edge gateway may report data to different service instances over a plurality of communication links, so that congestion on the communication link between the edge gateway and the internet of things platform can be avoided.

[0137] FIG. **6** is a flowchart of a method for sending a message by an internet of things platform to an edge gateway according to this application. Based on multi-link downlink communication, the internet of things platform actively sends a message (for example, delivers a control command) to the edge gateway. Refer to FIG. **6**, the method includes the following steps.

[0138] Step **601**: The edge gateway establishes a plurality of communication links with the internet of things platform.

[0139] Step **602**: A cluster management service on the internet of things platform sends a message (for example, a control command) to the edge gateway.

[0140] The cluster management service may be a service instance on the internet of things platform other than an access service and a load balancing service. The cluster management service may record a correspondence between the edge gateway and each service instance accessing the edge gateway.

[0141] Step **603**: The cluster management service randomly selects a service instance (for example, a service instance **1**) from service instances accessing the edge gateway to deliver the message.

[0142] Step **604**: LB sends the corresponding message to the edge gateway over a communication link corresponding to the service instance **1**.

[0143] Step **605**: After receiving a message on any communication link, the edge gateway normally processes a service request.

[0144] Step **606**: The cluster management service on the internet of things platform sends a message (for example, a control command) to the edge gateway.

[0145] Step **607**: The cluster management service randomly selects a service instance (for example, a service instance **2**) from service instances accessing the edge gateway to deliver the message.

[0146] Step **608**: The LB sends the corresponding message to the edge gateway over a communication link corresponding to the service instance **2**.

[0147] Step **609**: After receiving a message on any communication link, the edge gateway normally processes a service request.

[0148] In the method for sending a message by the internet of things platform to the edge gateway provided in this embodiment of this application, the internet of things platform may send a message to the edge gateway over a plurality of communication links, so that congestion on the communication link between the edge gateway and the internet of things platform can be avoided.

[0149] FIG. **7** is a flowchart of a method for disconnecting a connection to an edge gateway in a multi-link mode according to this application. Refer to FIG. **7**, the method includes the following steps.

[0150] Step **701**: The edge gateway disconnects communication links established with an internet of things platform in sequence.

[0151] Step **702**: The edge gateway disconnects a communication link **1**.

[0152] Step **703**: LB disconnects a communication link with a service instance **1**, and releases a binding relationship between the edge gateway corresponding to the communication link **1** and the service instance **1**.

[0153] Step **704**: The service instance **1** disconnects a link with the edge gateway, and releases the binding relationship between the edge gateway and the service instance **1**.

[0154] Step **705**: The edge gateway disconnects a communication link **2**.

[0155] Step **706**: The LB disconnects a communication link with a service instance **2**, and releases a binding relationship between the edge gateway corresponding to the communication link **2** and the service instance **2**.

[0156] Step **707**: The service instance **2** disconnects a link with the edge gateway, and releases the binding relationship between the edge gateway and the service instance **2**.

[0157] FIG. **8** is a diagram of a structure of a data transmission apparatus according to an embodiment of this application. The apparatus may be the edge gateway in the foregoing embodiment. As shown in FIG. **8**, the apparatus includes the following.

[0158] A link establishment module **810** configured to establish a first communication link with an internet of things platform, where the link establishment module **810** may be configured to implement link establishment functions corresponding to the foregoing step **201** and an implicit step of step **201**.

[0159] A transceiver module **820** configured to send a multi-link access request to the internet of things platform over the first communication link, where the multi-link access request carries a first quantity of internet of things devices accessing an edge gateway; and receive link establishment information that is sent by the internet of things platform and that corresponds to at least one second communication link, where the transceiver module **820** may be configured to implement receiving and sending functions corresponding to the foregoing step **202** and step **203** and implicit steps of step **202** and step **203**, where the link establishment module **810** is configured to establish the at least one second communication link with the internet of things platform based on the link establishment information corresponding to the at least one second communication link, and the link establishment module **810** may be configured to implement link establishment functions corresponding to the foregoing step **204** and an implicit step of step **204**.

[0160] A transmission module **830**, configured to perform data transmission with the internet of things platform over the first communication link and the second communication link, where the transmission module **830** may be configured to implement a transmission function corresponding to the foregoing step **205** and an implicit step of step **205**.

[0161] Optionally, the link establishment module **810** is configured to send, to the internet of things platform based on the link establishment information corresponding to the at least one second communication link, a link establishment request respectively corresponding to the at least one second communication link, where the link establishment request carries a gateway identifier of the edge gateway, the gateway identifier is different from a gateway identifier used by the edge gateway to establish the first communication link, and when there are a plurality of link establishment requests, each link establishment request carries a different gateway identifier.

[0162] Optionally, the link establishment information includes a private port number of a service instance corresponding to the second communication link. The link establishment module **810** is configured to send, to the internet of things platform, the link establishment request respectively corresponding to the at least one second communication link, where the link establishment request carries the private port number corresponding to the second communication link.

[0163] Optionally, the transmission module **830** is configured to select a target communication link from the first communication link and the second communication link according to a specified communication link selection policy each time data needs to be sent to the internet of things platform; and send the data over the target communication link.

[0164] Optionally, the communication link selection policy includes random selection or sequential selection.

[0165] Optionally, the multi-link access request further carries a communication frequency of the internet of things device.

[0166] An embodiment of this application further provides an edge gateway. The edge gateway includes a processor and a memory. The memory stores at least one instruction. The at least one instruction is loaded and executed by the processor to implement the data transmission method

provided in the foregoing embodiments. It should be noted that the link establishment module **810**, the transceiver module **820**, and the transmission module **830** may be implemented by the processor, or may be implemented by the processor in cooperation with components such as the memory.

[0167] FIG. **9** is a diagram of a structure of a data transmission apparatus according to an embodiment of this application. The apparatus may be the internet of things platform in the foregoing embodiment. As shown in FIG. **9**, the apparatus includes the following.

[0168] A transceiver module **910** configured to receive a multi-link access request sent by an edge gateway over a first communication link, where the multi-link access request carries a first quantity of internet of things devices accessing the edge gateway, and the transceiver module **910** may be configured to implement receiving and sending functions corresponding to the foregoing step **301** and an implicit step of step **301**.

[0169] A determining module **920** configured to determine, based on the first quantity, a second quantity of second communication links to be established with the edge gateway, where the determining module **920** may be configured to implement a determining function corresponding to the foregoing step **302** and an implicit step of step **302**.

[0170] An obtaining module **930** configured to obtain link establishment information corresponding to the second quantity of second communication links, where the obtaining module **930** may be configured to implement an obtaining function corresponding to the foregoing step **303** and an implicit step of the step **303**, where the transceiver module **910** is configured to send, to the edge gateway, the link establishment information corresponding to the second quantity of second communication links, and the link establishment information indicates the edge gateway to establish the second quantity of second communication links with the internet of things platform, and the transceiver module **910** may be configured to implement receiving and sending functions corresponding to the foregoing step **304** and an implicit step of step **304**.

[0171] A transmission module **940** configured to perform data transmission with the edge gateway over the first communication link and the second communication link, where the transmission module **940** may be configured to implement a transmission function corresponding to the foregoing step **305** and an implicit step of step **305**.

[0172] Optionally, the determining module **920** is configured to determine, based on a first correspondence, the second quantity that corresponds to the first quantity and that is of the second communication links to be established with the edge gateway.

[0173] Optionally, the multi-link access request further carries a communication frequency of the internet of things device accessing the edge gateway. The determining module **920** is configured to determine, based on the first quantity and the communication frequency, the second quantity of second communication links to be established with the edge gateway.

[0174] Optionally, the determining module **920** is configured to determine a product of the first quantity and the communication frequency; and determine, based on a second correspondence, the second quantity that corresponds to the product and that is of the second communication link to be established with the edge gateway.

[0175] Optionally, the obtaining module **930** is configured to determine, from service instances other than a service instance corresponding to the first communication link, service instances of the second quantity with least corresponding load; and obtain link establishment information respectively corresponding to the second quantity of service instances.

[0176] Optionally, the link establishment information includes a private port number corresponding to a service instance. The transceiver module **910** is further configured to receive a link establishment request that is sent by the edge gateway and that carries the private port number; and establish a second communication link between the edge gateway and a service instance corresponding to the private port number.

[0177] Optionally, the transmission module **940** is configured to select a target communication link

from the first communication link and the second communication link according to a specified communication link selection policy each time data needs to be sent to the edge gateway; and send the data over the target communication link.

[0178] Optionally, the communication link selection policy includes random selection or sequential selection.

[0179] The transceiver module **910**, the determining module **920**, the obtaining module **930**, and the transmission module **940** may all be implemented by using software, or may be implemented by using hardware. For example, the following uses the transceiver module **910** as an example to describe an implementation of the transceiver module **910**. Similarly, for implementations of the determining module **920**, the obtaining module **930**, and the transmission module **940**, refer to the implementation of the transceiver module **910**.

[0180] The module is used as an example of a software functional unit, and the transceiver module **910** may include code run on a computing instance. The computing instance may include at least one of a physical host (a computing device), a virtual machine, and a container. Further, there may be one or more computing instances. For example, the transceiver module **910** may include code run on a plurality of hosts/virtual machines/containers. It should be noted that, the plurality of hosts/virtual machines/containers configured to run the code may be distributed in a same region, or may be distributed in different regions. Further, the plurality of hosts/virtual machines/containers configured to run the code may be distributed in a same availability zone (AZ), or may be distributed in different AZs. Each AZ includes one data center or a plurality of data centers that are geographically close to each other. Generally, one region may include a plurality of AZs.

[0181] Similarly, the plurality of hosts/virtual machines/containers configured to run the code may be distributed in a same virtual private cloud (VPC), or may be distributed in a plurality of VPCs. Generally, one VPC is provided in one region. A communication gateway needs to be configured in each VPC for communication between two VPCs in the same region or between VPCs in different regions. Interconnection between VPCs is implemented through the communication gateway.

[0182] The module is used as an example of a hardware function unit, and the transceiver module **910** may include at least one computing device, for example, a server. Alternatively, the transceiver module **910** may be a device implemented by using an application-specific integrated circuit (ASIC) or a programmable logic device (PLD), or the like. The PLD may be implemented by a complex programmable logical device (CPLD), a field-programmable gate array (FPGA), generic array logic (GAL), or any combination thereof.

[0183] A plurality of computing devices included in the transceiver module **910** may be distributed in the same region, or may be distributed in different regions. The plurality of computing devices included in the transceiver module **910** may be distributed in the same AZ, or may be distributed in different AZs. Similarly, the plurality of computing devices included in the transceiver module **910** may be distributed in a same VPC, or may be distributed in a plurality of VPCs. The plurality of computing devices may be any combination of computing devices such as the server, the ASIC, the PLD, the CPLD, the FPGA, and the GAL.

[0184] It should be noted that, in another embodiment, the transceiver module **910** may be configured to perform any step in the data transmission method, the determining module **920** may be configured to perform any step in the data transmission method, and an obtaining module **930** may be configured to perform any step in the data transmission method. Steps implemented by the transceiver module **910**, the determining module **920**, the obtaining module **930**, and the transmission module **940** may be specified as required. The transceiver module **910**, the determining module **920**, the obtaining module **930**, and the transmission module **940** respectively implement different steps in the data transmission method, to implement all functions of the data transmission apparatus.

[0185] This application further provides a computing device **100**. The computing device **100** may be a server on the foregoing internet of things platform. As shown in FIG. **10**, the computing device

100 includes a bus **102**, a processor **104**, a memory **106**, and a communication interface **108**. The processor **104**, the memory **106**, and the communication interface **108** communicate with each other via the bus **102**. The computing device **100** may be a server or a terminal device. It should be understood that quantities of processors and memories in the computing device **100** are not limited in this application.

[0186] The bus **102** may be a Peripheral Component Interconnect (PCI) bus, an Extended Industry Standard Architecture (EISA) bus, or the like. The bus may include an address bus, a data bus, a control bus, and the like. For ease of representation, only one line is for representing the bus in FIG. **10**, but this does not mean that there is only one bus or only one type of bus. The bus **102** may include a channel for transferring information between various components (for example, the memory **106**, the processor **104**, and the communication interface **108**) of the computing device **100**.

[0187] The processor **104** may include any one or more of processors such as a central processing unit (CPU), a graphics processing unit (GPU), a microprocessor (MP), or a digital signal processor (DSP).

[0188] The memory **106** may include a volatile memory, for example, a random-access memory (RAM). The memory **106** may further include a non-volatile memory, for example, a read-only memory (ROM), a flash memory, a hard disk drive (HDD), or a solid-state drive (SSD).

[0189] The memory **106** stores executable program code. The processor **104** executes the executable program code to separately implement functions of the transceiver module **910**, the determining module **920**, the obtaining module **930**, and the transmission module **940**, to implement the data transmission method. In other words, the memory **106** stores instructions used for performing the data transmission method.

[0190] The communication interface **108** implements communication between the computing device **100** and another device or a communication network by using a transceiver module, for example, but not limited to a network interface card or a transceiver.

[0191] An embodiment of this application further provides a computing device cluster. The computing device cluster may be a server cluster on the foregoing internet of things platform, and the computing device cluster includes at least one computing device. The computing device may be a server, for example, a central server, an edge server, or a local server in a local data center. In some embodiments, the computing device may alternatively be a terminal device, such as a desktop computer, a notebook computer, or a smartphone.

[0192] As shown in FIG. **11**, the computing device cluster includes at least one computing device **100**. A memory **106** in the one or more computing devices **100** in the computing device cluster may store same instructions used for performing the foregoing data transmission method.

[0193] In some possible implementations, the memory **106** in the one or more computing devices **100** in the computing device cluster may alternatively store some instructions used for performing the foregoing data transmission method. In other words, a combination of the one or more computing devices **100** may jointly execute instructions used for performing the foregoing data transmission method.

[0194] It should be noted that memories **106** in different computing devices **100** in the computing device cluster may store different instructions respectively used for performing some functions of the data transmission apparatus. In other words, instructions stored in the memories **106** in different computing devices **100** may implement functions of one or more of the transceiver module **910**, the determining module **920**, the obtaining module **930**, and the transmission module **940**.

[0195] In some possible implementations, the one or more computing devices in the computing device cluster may be connected through a network. The network may be a wide area network, a local area network, or the like. FIG. **12** shows a possible implementation. As shown in FIG. **12**, two computing devices **100A** and **100B** are connected through a network. In an embodiment, communication interfaces in the computing devices are connected to the network. In this type of

possible implementation, a memory **106** in the computing device **100A** stores instructions used for performing a function of the transceiver module **910**. In addition, a memory **106** in the computing device **100B** stores instructions used for performing functions of the determining module **920**, the obtaining module **930**, and the transmission module **940**.

[0196] The connection manner between the computing device clusters shown in FIG. **12** may be that, considering that the data transmission method provided in this application requires a large computing amount, it is considered that the functions implemented by the determining module **920**, the obtaining module **930**, and the transmission module **940** are performed by the computing device **100B**.

[0197] It should be understood that functions of the computing device **100A** shown in FIG. **12** may alternatively be completed by the plurality of computing devices **100**. Similarly, functions of the computing device **100B** may alternatively be completed by the plurality of computing devices **100**.

[0198] An embodiment of this application further provides another computing device cluster. For a connection relationship between computing devices in the computing device cluster, refer to the connection manners of the computing device cluster in FIG. **11** and FIG. **12** similarly. A difference lies in that the memory **106** in the one or more computing devices **100** in the computing device cluster may store same instructions used for performing the foregoing data transmission method.

[0199] In some possible implementations, the memory **106** in the one or more computing devices **100** in the computing device cluster may alternatively store some instructions used for performing the foregoing data transmission method. In other words, a combination of the one or more computing devices **100** may jointly execute instructions used for performing the foregoing data transmission method.

[0200] An embodiment of this application further provides a data transmission system. The system includes an edge gateway and an internet of things platform. The edge gateway may implement the data transmission method implemented by the edge gateway in the foregoing embodiment. The internet of things platform may implement the data transmission method implemented by the internet of things platform in the foregoing embodiment.

[0201] An embodiment of this application further provides a computer program product including instructions. The computer program product may be software or a program product that includes the instructions and that can run on an edge gateway or that is stored in any usable medium. When the computer program product runs on the edge gateway, the edge gateway is enabled to perform the data transmission method.

[0202] An embodiment of this application further provides a computer program product including instructions. The computer program product may be software or a program product that includes instructions and that can run on a computing device or be stored in any usable medium. When the computer program product runs on at least one computing device, the at least one computing device is enabled to perform the foregoing data transmission method.

[0203] An embodiment of this application further provides a computer-readable storage medium. The computer-readable storage medium may be any usable medium that can be stored by a computing device, or a data storage device, for example, a data center, including one or more usable media. The usable medium may be a magnetic medium (for example, a floppy disk, a hard disk, or a magnetic tape), an optical medium (for example, a DIGITAL VERSATILE DISC (DVD)), a semiconductor medium (for example, a SSD), or the like. The computer-readable storage medium includes instructions. The instructions instruct an edge gateway to perform a data transmission method, or instruct a computing device to perform a data transmission method.

[0204] Finally, it should be noted that the foregoing embodiments are merely intended for describing the technical solutions of this application, but not for limiting this application. Although this application is described in detail with reference to the foregoing embodiments, a person of ordinary skill in the art should understand that modifications may still be made to the technical solutions described in the foregoing embodiments or equivalent replacements may be made to

some technical features thereof, without departing from the protection scope of the technical solutions in embodiments of this application.

[0205] The foregoing descriptions are merely embodiments of this application, but are not intended to limit this application. Any modification, equivalent replacement, or improvement made within the principle of this application should fall within the protection scope of this application.

Claims

1. A method implemented by an internet of things platform, wherein the method comprises: receiving, from an edge gateway via a first communication link, a multi-link access request comprising a first quantity of internet of things devices accessing the edge gateway; determining, based on the first quantity, a second quantity of second communication links to be established with the edge gateway; obtaining link establishment information corresponding to the second quantity; sending, to the edge gateway, the link establishment information to indicate the edge gateway to establish the of second communication links with the internet of things platform; and performing data transmission with the edge gateway over the first communication link and the second communication links.
2. The method of claim 1, wherein determining the second quantity comprises determining, based on a first correspondence, the second quantity.
3. The method of claim 1, wherein the multi-link access request further carries a communication frequency of the internet of things devices accessing the edge gateway, and wherein determining the second quantity comprises determining, based on the communication frequency, the second quantity.
4. The method of claim 3, wherein determining the second quantity further comprises: determining a product of the first quantity and the communication frequency; and determining, based on a second correspondence, the second quantity that corresponds to the product.
5. The method of claim 1, wherein obtaining the link establishment information comprises: determining, from among first service instances other than a service instance corresponding to the first communication link, second service instances equal to the second quantity and having the least load; and obtaining link establishment information respectively corresponding to the second service instances.
6. The method of claim 1, wherein before performing the data transmission, the method further comprises: receiving a link establishment request that is from the edge gateway and that carries a private port number; and establishing a third communication link between the edge gateway and a service instance corresponding to the private port number.
7. The method of claim 1, wherein performing the data transmission comprises: selecting a target communication link from among the first communication link and the second communication links and according to a communication link selection policy each time before sending data to the edge gateway; and sending the data over the target communication link.
8. The method of claim 7, wherein the communication link selection policy comprises random selection or sequential selection.
9. An apparatus comprising: a memory configured to store instructions; and one or more processors coupled to the memory, wherein when executed by the one or more processors, the instructions cause the apparatus to: receive, from an edge gateway via a first communication link, a multi-link access request comprising a first quantity of internet of things devices accessing the edge gateway; determine, based on the first quantity, a second quantity of second communication links to be established with the edge gateway; obtain link establishment information corresponding to the second quantity; send, to the edge gateway, the link establishment information to indicate the edge gateway to establish the second communication links with the data transmission apparatus; and perform data transmission with the edge gateway over the first communication link and the second

communication links.

10. The apparatus of claim 9, wherein when executed by the one or more processors, the instructions further cause the apparatus to determine the second quantity by determining, based on a first correspondence, the second quantity.

11. The apparatus of claim 9, wherein the multi-link access request further carries a communication frequency of the internet of things device accessing the edge gateway, and wherein when executed by the one or more processors, the instructions further cause the apparatus to determine the second quantity by determining, based on the communication frequency, the second quantity of the second communication links.

12. The apparatus of claim 11, wherein when executed by the one or more processors, the instructions further cause the apparatus to determine the second quantity by: determining a product of the first quantity and the communication frequency; and determining, based on a second correspondence, the second quantity that corresponds to the product.

13. The apparatus of claim 9, wherein when executed by the one or more processors, the instructions further cause the apparatus to obtain the link establishment information by: determining, from first among service instances other than a service instance corresponding to the first communication link, second service instances equal to the second quantity and having the least load; and obtaining link establishment information respectively corresponding to the second service instances.

14. The apparatus of claim 9, wherein before performing the data transmission, when executed by the one or more processors, the instructions further cause the apparatus; to: receive a link establishment request that is from the edge gateway and that carries a private port number; and establish a third communication link between the edge gateway and a service instance corresponding to the private port number.

15. The apparatus of claim 9, wherein when executed by the one or more processors, the instructions further cause the apparatus to perform the data transmission by: selecting a target communication link from among the first communication link and the second communication links and according to a communication link selection policy each time before sending data to the edge gateway; and sending the data over the target communication link.

16. The apparatus of claim 15, wherein the communication link selection policy comprises random selection or sequential selection.

17. A computer program product comprising computer-executable instructions that are stored on a non-transitory computer-readable storage medium and that, when executed by one or more processors, cause an apparatus to: receive, from an edge gateway via a first communication link, a multi-link access request carrying a first quantity of internet of things devices accessing the edge gateway; determine, based on the first quantity, a second quantity of second communication links to be established with the edge gateway; obtain link establishment information corresponding to the second quantity of the second communication links; send, to the edge gateway, the link establishment information indicating the edge gateway to establish the second quantity of second communication links with the data transmission apparatus; and perform data transmission with the edge gateway over the first communication link and the second communication link.

18. The computer program product of claim 17, wherein the instructions that, when executed by the one or more processors, cause the apparatus to determine the second quantity by determining, based on a first correspondence, the second quantity.

19. The computer program product of claim 17, wherein the multi-link access request further carries a communication frequency of the internet of things device accessing the edge gateway, wherein the instructions that, when executed by the one or more processors, cause the apparatus to determine the second quantity by determining, based on the communication frequency, the second quantity.

20. The computer program product of claim 19, wherein the instructions that, when executed by the

one or more processors, cause the apparatus to determine the second quantity by: determining a product of the first quantity and the communication frequency; and determining, based on a second correspondence, the second quantity that corresponds to the product.
